

Co-Design MCP



Overview

The Co-Design extension is a suite of in-house skills and MCP (Model Context Protocol) tools developed by the AI Solutions Group (AISG) to enable structured interaction between VS Code, GHCP, AI assistants, and the capabilities available in Co-Design.

By integrating GHCP, an interactive AI agent chat, with Intel's design and engineering knowledge—including project specifications (HAS/MAS/FAS), tools, methodologies, HSD tenants, remote codebases, and advanced debug and validation tools and skills—Co-Design allows you to handle various tasks end-to-end, directly within VS Code using GitHub Copilot, in addition to standard Copilot capabilities.

Co-Design standardizes how questions are interpreted, projects are selected, and tools are invoked, ensuring answers are based on validated data.

This wiki provides guidance on using Co-Design MCP in VS Code, from setup to daily workflows and expected behaviors.

Co-Design MCP is continuously evolving, with new features and updates added regularly- this page will be updated as new capabilities become available.

Want to learn more about MCPs?

[Click here to read more](#)

MCP Tools

The Co-Design MCP consists of several tools:

Co-Design MCP For Specs and Wikis

1. **codesign-get-spec-sources** - Returns the list of projects currently available to the user for querying. The same projects that the user has access to in the Co-Design Chat UI. (Click [here](#) to view the full access list)
2. **codesign-ask-specs-and-wikis** - Queries specification and tools & methodologies content within identified projects to answer the user's question.

Co-Design MCP For HSD

codesign-ask-hsd-agent - Queries HSD tickets content within identified tenants subjects to answer the user's question.

Co-Design Debug MCP

Orchestrator Tool:

codesign-debug - Orchestrates a full debug workflow:

- Extract failure data from RPT files
- Searches and retrieves relevant debug memories
- Fetches vManager metadata
- Guides investigation through logs, specs, code and users' custom debug instructions
- Stores solutions as reusable memories
- Creates and auto-populates HSD tickets

Debug Tools:

Available for direct use, integration in existing agents\workflows or invoked automatically by codebug-debug

- **codesign-debug-get-bucket-metadata-from-vmanager** - Retrieves bucket metadata from vManager including debug status, failure notes, ownership, first occurrence, and associated bug IDs.

- **codesign-debug-list-available-memories** - Lists all stored debugging sessions for the current bucket and cluster-stepping.
- **codesign-debug-search-in-memories** - Searches keywords across stored memories to find relevant past investigations.
- **codesign-debug-get-memory** - Retrieves details of a specific memory including full action items and methodology.
- **codesign-debug-store-memory** - Stores your debugging session with symptoms, root cause, action items, and tags for team reuse.
- **codesign-debug-get-extra-hsd-fields-schema** - Gets the schema of additional required fields to create an HSD ticket for a specific cluster/stepping.
- **codesign-debug-create-hsd-ticket** - Auto-populates HSD ticket fields based on debugging context and returns a review link for you to finalize and submit.

FSDB Analysis Tools

These are local tools that are available in EC environments, enabling waveform-level debugging directly through the Copilot Agent.

- **codesign-debug-get-fsdb-usage-instructions** tool provides Copilot with guidance on how to interpret FSDB files, with a focus on understanding signal transitions and effectively using CoDesign FSDB analysis tools.
- **codesign-debug-run-npi-analysis-on-fsdb** tool delivers core querying capabilities for waveform data. It supports retrieving signal values at a specific point in time, extracting signal transitions over a defined time window, and dumping the FSDB hierarchy to help map signals between CTE and RTL (with optional scoping).

Together, these tools allow the agent to programmatically access and analyze waveform data, enabling automated signal investigation as part of the debug workflow and reducing the need for manual waveform inspection.

Ask Bucket Skill and Tools

Ask Bucket is a bucket-level investigation capability that enables engineers to ask questions across all failures in a selected bucket, rather than analyzing a single failure in isolation.

- **codesign-debug-ask-bucket** - Skill that orchestrate the bucket wide investigation:
 - Defines the question
 - Retrieve all the bucket failures paths from vManager
 - Solves the question for a sample failure
 - Translates the sample solution into patterns and builds the bucket-wide plan
 - Runs the pattern search across the whole bucket
 - Analyzes the output and synthesize the answer
- **codesign-debug-get-bucket-failure-paths** - a tool that fetches all the failures paths of the bucket, model version and project from vManager.
- **codesign-debug-search-bucket-patterns** - a **local tool** that runs regex patterns across all failures artifacts and returns matches that can be counted, grouped, and summarized.

Together, these skill and tools allow the agent to quickly understand the scope, identify root causes, and reach confident data-driven conclusions - all within VS Code.

Co-Design MCP For Code

1. **codesign-get-remote-code-sources** - Returns the list of code projects and IP scopes currently available to the user for querying. These are the same sources available in the CoDesign Chat UI.
1. **codesign-ask-remote-code-repo** - Queries source code content within the identified IP repositories to deliver highquality, codegrounded answers

Co-Design MCP For ValPlan AI

1. **codesign-valplanai skill** -
 - Guides you through a phased navigation flow: plan list plan overview feature details TCD list TCD test cases.
 - Shows sources when available, including clickable document links when provided.
 - Supports answering your questions by first searching inside the retrieved ValPlan content, then using related source documents in Code chat ask tool.
2. **codesign-valplanai-get-list** - List your 20 most recent validation plans from ValPlan AI and assist with relevant plan selection.
3. **codesign-valplanai-get-overview** - Retrieves an overview of a selected ValPlan AI-generated plan, including plan metadata, feature list, and statuses, source documents, and TCD counts.
4. **codesign-valplanai-get-feature** - Retrieves detailed information for a selected feature generated by ValPlan AI, including description, status, sources, and the list of related TCDs.
5. **codesign-valplanai-get-tcd** - Retrieves the full details of a selected TCD, including title, description, sources, and test attributes categorized by type.

Contact us

We're Here for You, Every Step of the Way.

Have a question, need assistance, or want to share feedback?

We are here to answer any questions you might have.

Connect with us through your preferred channel:

Email support

For direct, private, or sensitive inquiries, email us at co-design-mcps@intel.com

Yammer Community

Join our vibrant Yammer community at go.to/codesign_yammer to ask questions and get peer support.

Ready to Dive In?

Start your journey with Co-Design MCP by browsing the sections below.